Trigonometric Approach to Inverse Kinematics

Summary:

This lesson introduces inverse kinematics (IK): the reverse of the forward kinematics from Lessons 3–5. Forward kinematics turns joint angles into a tip position. Inverse kinematics starts from a target position and finds the joint angles. On Day 1, students solve the flat arm with triangles: the two links and the line D from the shoulder to the target make a triangle, so the law of cosines gives the elbow angle θE and the shoulder angle θS. On Day 2, students add the base turn: atan2 aims the base, the target is flattened into the arm's upright plane, and Day 1's steps finish the job. Every answer is checked with forward kinematics, using the link table, the Desmos graph from Lesson 4 or the code from Lesson 5. All angles use the DH convention from Lessons 4 and 5.

Objective:

By the end of this lesson, students will be able to:

- Explain the difference between forward and inverse kinematics
- Use the law of cosines to find an angle of a triangle when all three sides are known
- Explain why the two links and the line D to the target make a triangle, and use it to find θE and θS for the flat arm
- Find the base angle θB with atan2, and flatten a 3D target into the arm's plane with $R = \sqrt{(x^2 + y^2)}$
- Decide whether a target is reachable, and explain why a target can have two answers (elbow up and elbow down)
- Check an IK answer with forward kinematics, and describe the strengths and limits of the trigonometric method

Standards:

NGSS

- Engineering Design: HS-ETS1-2 (break a complex problem into smaller problems), HS-ETS1-3 (evaluate solutions against trade-offs), HS-ETS1-4 (use a computer model to test a solution)
- Science & Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking; Engaging in Argument from Evidence
- Crosscutting Concepts: Systems and System Models; Patterns

Vocabulary:

Inverse kinematics (IK): Finding the joint angles that put the tip of the arm on a target. It is the reverse of forward kinematics (FK).

Target: The position we want the tip to reach: (X, Y) for the flat arm and (x, y, z) in 3D.

Law of cosines: A rule for any triangle: $c^2 = a^2 + b^2 - 2ab \cos C$. Solved for the angle, $\cos C = (a^2 + b^2 - c^2) \div (2ab)$. It finds an angle when all three sides are known.

Inverse cosine ($\cos^{-1}$): The calculator key (also called arccos) that turns a cosine back into its angle. Use degrees mode. In Python it is math.acos, which works in radians.

Inverse tangent ($\tan^{-1}$): The calculator key (also called arctan) that turns a tangent back into its angle. It cannot tell some directions apart, so the base uses atan2.

D: The straight-line distance from the shoulder to the target. For the flat arm $D = \sqrt{(X^2 + Y^2)}$. D and the two links make a triangle.

γ (gamma): The angle inside the triangle at the elbow. The elbow bend θE is the outside angle: its size is $180° - \gamma$.

φ and δ (phi and delta): The two parts of the shoulder angle. $\varphi = \tan^{-1}(Y \div X)$ is the tilt of the line D. δ is the angle between D and the upper arm, found with the law of cosines. Elbow up, $\theta S = \varphi + \delta$.

atan2(y, x): A version of $\tan^{-1}$ that looks at the signs of both y and x, so it always points the right way. It gives the base angle θB.

Reach R: How far the target is from the base axis across the floor: $R = \sqrt{(x^2 + y^2)}$. It is the same R as in Lesson 4.

Elbow up / elbow down: The two arm shapes that usually reach the same target. Elbow up (our usual shape) has θE negative; elbow down has θE positive.

Reachable: A target the arm can reach: $D \leq L1 + L2 = 500$ mm. If cos θE comes out bigger than 1 or less than −1, the target is out of reach.

Resources & Materials:

- Lesson 6 slideshow, Days 1 and 2 (37 slides; the slide numbers in the Procedures are from this slideshow)
- Calculators with $\cos^{-1}$ and $\tan^{-1}$ keys, set to degrees mode, for the Try It Yourself problems
- Computers with Python 3 and arm.py from Lesson 5, for checking answers with tip_3d (optional). No extra libraries are needed
- The Desmos graph from Lesson 4 and students' link tables from Lesson 4, for checking answers with forward kinematics
- The assembled 4-DOF arm from Lesson 2, to show the reach and the elbow-up and elbow-down shapes (optional)
- All numbers use L1 = L2 = 250 mm and the DH angles from Lessons 4 and 5: θB is the base turn, θS is measured from the floor, θE is measured from the line of the upper arm, and up is +.

Procedures:

Day 1: The Flat Arm with Triangles

- **Opening (5 min)** — Show the title slide, "Where Lesson 5 Left Off" and "Forward vs. Inverse Kinematics" (slides 1–3). In Lesson 5, `tip_3d(45, 45, -45)` turned the angles θB = 45°, θS = 45°, θE = −45° into the tip (301.78, 301.78, 176.78). Today we go the other way: a robot is told a

target, like “pick up the cup”, and its motors need angles. FK always has exactly one answer; IK can have no answer, one or two.

- **The IK problem (3 min)** — Show “The IK Problem (Flat Arm)” (slide 4). We know L1 = L2 = 250 mm and the target (X, Y), and we want θS and θE. Solve the flat arm first; the base turn is added on Day 2. Remind students of the DH angles: θS is measured from the floor, θE from the line of the upper arm, and up is +.
- **The law of cosines (7 min)** — Show “The Law of Cosines” (slide 5): $c^2 = a^2 + b^2 - 2ab \cos C$, and solved for the angle, $\cos C = (a^2 + b^2 - c^2) \div (2ab)$. Point out that C = 90° gives cos C = 0, which is the Pythagorean theorem. Work the 250, 250, 300 triangle on “Worked Example: The Law of Cosines” (slide 6): cos C = 0.28, so C = 73.74°. Check that calculators are in degrees mode.
- **Try It Yourself #1 (5 min)** — Students find the angle C for sides 250, 250, 250 and 250, 250, 400 (slide 7). Reveal the solutions 60° and 106.26° (slide 8). Use (b) to show that a negative cosine gives an angle bigger than 90°.
- **The arm makes a triangle (4 min)** — Show “The Arm Makes a Triangle” (slide 9). Draw the line D from the shoulder to the target. The two links and D make a triangle with three known sides: L1, L2 and $D = \sqrt{(X^2 + Y^2)}$. So the law of cosines can find its angles, and those angles give the joint angles.
- **Step 1: the elbow (9 min)** — Show “Step 1: The Elbow Angle θE” (slide 10). The law of cosines gives γ, the angle inside the triangle at the elbow. θE is the bend away from a straight arm, so its size is 180° − γ. Because cos(180° − γ) = −cos γ, this becomes $\cos \theta E = (D^2 - L1^2 - L2^2) \div (2\ L1\ L2)$. Check it with a straight arm: D = 500 gives cos θE = 1 and θE = 0°. Our forearm bends down (elbow up), so θE is negative. Then work “Worked Example: The Elbow” (slide 11) for (426.78, 176.78): D^2 = 213,392, cos θE = 0.7071, so θE = −45°. Common mistake: using γ = 135° as the elbow angle.
- **Step 2: the shoulder (8 min)** — Show “Step 2: The Shoulder Angle θS” (slide 12). θS is measured from the floor and is made of two angles: $\varphi = \tan^{-1}(Y \div X)$, the tilt of the line D, and δ, the angle between D and the upper arm, from $\cos \delta = (L1^2 + D^2 - L2^2) \div (2\ L1\ D)$. Because L1 = L2, this is just cos δ = D ÷ 500. Elbow up, the upper arm is above D, so θS = φ + δ. No conversion is needed, because DH measures θS from the floor. Work “Worked Example: The Shoulder” (slide 13): φ = 22.5°, δ = 22.5°, θS = 45°.
- **The answer (3 min)** — Show “The Answer” (slide 14). IK found θS = 45° and θE = −45°, the angles Lesson 4 started with. Check with FK: `show(flat_tip(45, -45))` prints [426.78, 176.78], back on the target.
- **Try It Yourself #2 and #3 (8 min)** — Students solve the targets (341.51, 341.51) and (250, 250), elbow up (slides 15 and 17). Reveal θS = 60°, θE = −30° (slide 16) and θS = 90°, θE = −90° (slide 18). The first is Lesson 4’s Try It Yourself #2 run backwards, so students can compare with their link tables. In the second, the upper arm points straight up and the forearm points straight across.
- **Elbow up or elbow down (3 min)** — Show “Elbow Up or Elbow Down” (slide 19). $\cos^{-1}$ only gives a positive angle, so we choose the sign. Elbow up: θE = −45° and θS = φ + δ = 45°. Elbow down: θE = +45° and θS = φ − δ = 0°. Both put the tip on (426.78, 176.78). This course uses elbow up. Teacher note: check the real arm’s joint limits before using elbow down. If the arm is in the room, show both shapes.

- **Can the arm reach it? (3 min)** — Show "Can the Arm Reach It?" (slide 20). A triangle can only close if $D \leq L1 + L2 = 500$ mm. If D is bigger, cos θE comes out bigger than 1, and $\cos^{-1}$ gives an error because no angle has a cosine bigger than 1. Always check cos θE first. If it is not between −1 and 1, the target is out of reach and the robot must not move.
- **Recap (2 min)** — Show "Day 1 Recap" (slide 21): the triangle, the elbow, the shoulder and the checks. Preview Day 2: the real arm's base turns too, so we need 3D.

Day 2: Adding the Base Turn (3D)

- **Recap (3 min)** — Show "Day 2: Where Day 1 Left Off" (slide 22). Ask: What is θE for a straight arm? (0°.) What is the largest D the arm can reach? (500 mm.) What is the sign of θE for elbow up? (Negative.)
- **Step 1: the base aims the arm (4 min)** — Show "Step 1: The Base Aims the Arm" (slide 23). From above, the base turns until the arm points at the target's (x, y) spot. Lesson 4 used $x = R \cos \theta B$ and $y = R \sin \theta B$, so $y \div x = \tan \theta B$ and $\theta B = \tan^{-1}(y \div x)$. For (301.78, 301.78, 176.78), $\theta B = \tan^{-1}(1) = 45°$.
- **Step 2: flatten 3D into 2D (4 min)** — Show "Step 2: Flatten 3D Into 2D" (slide 24). The arm always stays in one upright plane, the plane the base points at. Inside that plane, the target is $R = \sqrt{(x^2 + y^2)}$ out and z up. Then (R, z) plays the role of (X, Y) from Day 1, and the elbow and shoulder steps do not change.
- **Why atan2? (5 min)** — Show "Why atan2?" (slide 25). For the spot (−125, 216.51), $\tan^{-1}(216.51 \div -125) = -60°$, which points the wrong way, while atan2(216.51, −125) = 120°. $y \div x$ is the same for (−125, 216.51) and (125, −216.51), so $\tan^{-1}$ cannot tell them apart; atan2 looks at the signs of both. In Python: `math.degrees(math.atan2(y, x))`. Teacher note: atan2 gives angles from −180° to 180°. If x = y = 0, the target is right above the base and any θB works.
- **The full recipe (3 min)** — Show "The Full Recipe" (slide 26): (1) θB = atan2(y, x); (2) $R = \sqrt{(x^2 + y^2)}$ and $D^2 = R^2 + z^2$; (3) $\cos \theta E = (D^2 - L1^2 - L2^2) \div (2\ L1\ L2)$, after checking that it is between −1 and 1, and θE is negative for elbow up; (4) $\varphi = \tan^{-1}(z \div R)$, $\cos \delta = (L1^2 + D^2 - L2^2) \div (2\ L1\ D)$ and $\theta S = \varphi + \delta$.
- **Worked example (5 min)** — Walk through the target (301.78, 301.78, 176.78) on "Worked Example" (slide 27). θB = 45°, R = 426.78 and z = 176.78. This is Day 1's worked example, so θE = −45° and θS = 45°. Ask students to predict the answer first: the point came from Lesson 4, so it should be 45°, 45°, −45°.
- **Check it with FK (4 min)** — Show "Check It with FK" (slide 28). Put the IK angles back into FK and see that the tip lands on the target. There are three ways: Lesson 4's link table by hand, the Desmos graph (u = 45, v = 45, w = −45 gives d = (3.0178, 3.0178, 1.7678), which is (301.78, 301.78, 176.78) mm), and Lesson 5's `show(tip_3d(45, 45, -45))`. Make it a habit: always check IK with FK.
- **Try It Yourself #4 (8 min)** — Students solve (295.75, 170.75, 91.51) (slide 29). Reveal θB = 30°, θS = 60°, θE = −90° (slide 30). Here R = 341.50, D = 353.55 and cos θE = 0. This is Lesson 4's Try It Yourself #5 run backwards. Have students check each answer from #4 to #6 with FK: the link table, the Desmos graph or tip_3d from Lesson 5.
- **Try It Yourself #5 (7 min)** — Students solve (−324.76, 187.50, 216.51) (slide 31). Reveal θB = 150°, θS = 60°, θE = −60° (slide 32). Plain $\tan^{-1}$ would give −30°. If time is short, assign this one as homework.

- **Try It Yourself #6 (7 min)** — Students decide whether (200, 200, 200) and (400, 300, 200) are reachable (slide 33). Reveal (slide 34): D = 346.41 mm, reachable; D = 538.52 mm, out of reach ($\cos \theta E = 1.32$). For an unreachable target the robot should report an error instead of moving.
- **When IK gets tricky (4 min)** — Show "When IK Gets Tricky" (slide 35). FK always gives one answer; IK can give none (D > 500 mm), one (D = 500 mm, a straight arm), two (elbow up or down), or infinitely many (x = y = 0). Near full reach, rounding matters: (433.01, 250.00) is really a straight arm, but the rounded numbers give $\theta E = -0.35°$. If rounding pushes the cosine just above 1 (like 1.001), treat it as 1: the arm is straight, not out of reach.
- **Trigonometry for IK (3 min)** — Show "Trigonometry for IK: Pros and Cons" (slide 36). Ask: if the arm had six joints, which method would you use, and why? Trigonometry gives an exact, fast answer for our arm, but every new joint needs new triangles and special cases.
- **Wrap-up (3 min)** — Show "What We Learned" (slide 37): IK goes from a target to the angles; the arm makes a triangle, so the law of cosines gives θE and δ; $\theta B = \text{atan2}(y, x)$, and R flattens 3D into 2D; $\theta S = \varphi + \delta$; always check with FK. Preview Lesson 7: gradient descent, a method that works for any arm.

Assessment:

Students complete the Try It Yourself problems (#1–#6) and check each answer against the solution on the slide. They can also check #4 to #6 with forward kinematics: the link table, the Desmos graph or tip_3d from Lesson 5.

Use these questions as an exit ticket or short quiz. The answers are in red.

- A triangle has sides 250, 250 and 350. Find the angle opposite the side of length 350.
  Answer: $\cos C = (62{,}500 + 62{,}500 - 122{,}500) \div 125{,}000 = 0.02$, so $C = \cos^{-1}(0.02) = 88.85°$.

- What is the difference between forward and inverse kinematics?
  Answer: FK goes from the joint angles to the tip position and always has one answer. IK goes from a target position to the joint angles and can have no answer, one or two.

- The arm has L1 = L2 = 250 mm. Can it reach (400, 300, 200)? Show why or why not.
  Answer: No. $D = \sqrt{(400^2 + 300^2 + 200^2)} = 538.52$ mm, which is more than 500 mm. $\cos \theta E$ would be 1.32, which is impossible.

- Find θS and θE (elbow up) for the flat-arm target (0, 500).
  Answer: D = 500, so $\cos \theta E = 1$ and $\theta E = 0°$. $\varphi = 90°$ and $\cos \delta = 500 \div 500 = 1$, so $\delta = 0°$ and $\theta S = 90°$. The arm points straight up.

- Why do we use atan2(y, x) instead of $\tan^{-1}(y \div x)$ for the base angle θB?
  Answer: $\tan^{-1}$ cannot tell opposite directions apart (and fails when x = 0). atan2 uses the signs of both y and x, so it gives the right angle, such as 120° for (−125, 216.51).

- True or false: every target the arm can reach has exactly one set of joint angles.
  Answer: False. Most targets have two (elbow up and elbow down). A fully straight arm (D = 500 mm) has one, and a target right above the base has infinitely many values of θB.

- List the four steps of the trigonometric IK recipe.

Answer: (1) $\theta B = \text{atan2}(y, x)$. (2) $R = \sqrt{(x^2 + y^2)}$ and $D^2 = R^2 + z^2$. (3) $\cos \theta E = (D^2 - L1^2 - L2^2) \div (2\ L1\ L2)$, negative for elbow up. (4) $\theta S = \varphi + \delta$, with $\varphi = \tan^{-1}(z \div R)$ and $\cos \delta = (L1^2 + D^2 - L2^2) \div (2\ L1\ D)$.

- Find θB, θS and θE (elbow up) for the target (0, 250, 250), and check it.
  Answer: $\theta B = \text{atan2}(250, 0) = 90°$. $R = 250$ and $z = 250$, so $D^2 = 125{,}000$, $\cos \theta E = 0$ and $\theta E = -90°$. $\varphi = 45°$ and $\delta = 45°$, so $\theta S = 90°$. Check: tip_3d(90, 90, -90) gives [0.0, 250.0, 250.0].

- In one or two sentences, why is trigonometry a good choice for our arm but not for a six-joint arm?
  Answer: For our arm the triangles give an exact, fast answer. A six-joint arm needs new triangles and many special cases for every extra joint, so a method that does not depend on drawing triangles works better.